Matrices & Matrix Algebra

Matrix addition and scalar multiplication

1 Let A have rows $\langle 1, 2\rangle, \langle 3, 4\rangle$ and B have rows $\langle 0, -1\rangle, \langle 2, 1\rangle$. Find $A + B$ and $2A - B$.

Matrix multiplication

2 Let A have rows $\langle 1, 2\rangle, \langle 3, 4\rangle$ and B have rows $\langle 2, 0\rangle, \langle 1, 3\rangle$. Find AB.

Matrix multiplication is not commutative

3 For A, B above, compute BA and compare it to AB.

The identity matrix and matrix inverses

4 Describe the 2×2 and 3×3 identity matrices by their rows.

Invertibility of 2x2 matrices

5 For A with rows $\langle a, b\rangle, \langle c, d\rangle$, state the formula for A^{-1}.

6 Find the inverse of A with rows $\langle 3, 1\rangle, \langle 2, 1\rangle$.

Inverse of a product

7 State the formula for $(AB)^{-1}$ in terms of A^{-1} and B^{-1}.

8 If A^{-1} has rows $\langle 2, 0\rangle, \langle 1, 1\rangle$ and B^{-1} has rows $\langle 1, 1\rangle, \langle 0, 1\rangle$, find $(AB)^{-1}$.